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Original Research

MAM: Flexible Monte-Carlo Agent based model for modelling COVID-19 spread

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ABSTRACT

In the three years since SARS-CoV-2 was first detected in China, hundreds of millions of people have been infected and millions have died. Along with the immediate need for treatment solutions, the COVID-19 epidemic has reinforced the need for mathematical models that can predict the spread of the pandemic in an ever-changing environment. The susceptible-infectious-removed (SIR) model has been widely used to model COVID-19 transmission, however, with limited success. Here, we present a novel, dynamic Monte-Carlo Agent-based Model (MAM), which is based on the basic principles of statistical physics. Using public aggregative data from Israel on three major outbreaks, we compare predictions made by SIR and MAM, and show that MAM outperforms SIR in all aspects. Furthermore, MAM is a flexible model and allows to accurately examine the effects of vaccinations in different subgroups, and the effects of the introduction of new variants.

1. Introduction

In the 18th century, Swiss mathematician Daniel Bernoulli developed mathematical models to study how variolation could be used to control smallpox [1]. Since then, researchers have used many approaches to develop models that can examine and explore the dynamics of infectious disease transmission. Nowadays, mathematical modelling of disease spread has become essential for decision-making on the national and international level, especially during the COVID-19 pandemic [2-5]. Globalization had, however, changed geography, social conditions, and transportation across countries, even compared to twenty years ago when SARS-CoV-1 was first detected. Consequently, mathematical models need to be adapted to a world that is not static but varies from place to place and over time.

There are three main approaches to model pandemic spread: empirical, mathematical, and individual models [6-9]. Empirical, or statistical, models use available data and statistical and machine-learning-based approaches to make predictions on future spread dynamics or the temporal evolution of the disease in terms of severe morbidity. The main drawback of such models in COVID-19 is their inability to predict future outbreaks in the presence of changing conditions, for example, a scenario of a new variant or the vaccination of the population. Statistical models, however, had a major role in clinical outcome predictions of COVID-19 patients since disease progression was associated with individuals' health conditions more than with the variant [10].

Mathematical models are sets of coupled differential equations to predict the spread of disease. The SIR methodology [11] and its refinements, such as the SEIR model [12] are the main known examples of this class. They have been the dominant approach in the scientific literature for studying infectious diseases and have been applied widely over the last decades. In general, SEIR stands for "Susceptible, Exposed, Infected, Removed (or Recovered)" which serves to decompose the population. The model describes a pandemic via the movement of the population from one compartment to another. Susceptible individuals become exposed, infected, and finally removed from the population. Removal can be either by recovery or by death. Transitions from one state to another are governed by set of rate equations. Since the introduction of SARS-CoV-2, SIR and SEIR have been heavily utilized to study the dynamics of the pandemic [13-18].

One disadvantage of population-level mathematical models is that they are limited in the ability to model system dynamics, especially when various population subgroups have different dynamics. We argue that a model that can accurately predict the spread must be able to model subpopulations that may have different spread dynamics such that each subgroup needs the have its own set of equations. For example,

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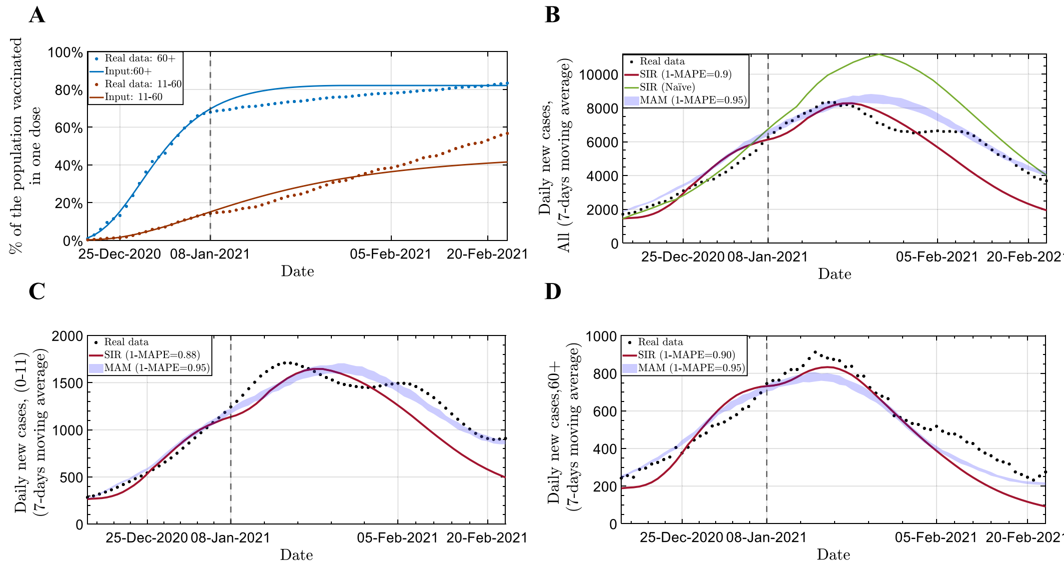

Fig. 1. Modelling the outbreak spread in Israel during the first vaccination campaign. A. Vaccination rates for two age groups based on real data up to January 8th, 2021, and predicted rates from this day forward. Actual vaccination rates are in dots. **B-D**. Predictions for the number of confirmed cases in Israel from Dec 25th, 2020, until Feb 20th, 2021 (7-days moving average) for the whole population (**B**), ages 0–11, which are all unvaccinated currently (**C**); and for 60 + years old (**D**). For panels B-D: real data in dots, SIR predictions in solid red line and MAM predictions in blue band (with 95% prediction interval). Dashed black line on January 8^{tt}, 2021, shows the index date from which predictions begin. Green line in **B** is a prediction made with SIR without breaking down age groups. MAPE statistics are presented in the legend. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

since nonpharmaceutical interventions (NPIs) are essential in controlling COVID-19 and they tend to fluctuate across regions and countries, the model must be able to distinguish regionality. Similarly, different age groups have varying patterns of social interactions, and in accordance, the model is required to distinguish age. Creating a mathematical model that models subpopulations, however, would require writing separate equations for each subpopulation, which would make the model highly complex.

Agent-based models (ABM) representing individuals separately can be an effective alternative to mathematical models in dealing with these issues [19-23]. Each individual can be represented by a particle, and interaction between particles can represent social interactions. Microsimulation modeling approaches can then be used with rules governing those interactions depending on the characteristics of the particles (such age and vaccination status). This model allows all the individuals to behave and interact according to separate characteristics and we can than observe the resulting macroscopic and microscopic impact on society.

In 2018, a study by Hunter et al. [9] compared agent-based (ABM) and equation-based models for predicting the overall number of cases in an outbreak of an infectious disease by simulating measles outbreaks in 33 Irish towns. In this study, agents representing individuals move between their current location and desired destination in a straight line, interacting with other agents along the way. The study used data from the Irish Central Statistics Offices on vaccination rates by age to predict the spread of measles outbreaks in each town. According to the study, although ABM requires more effort to set up and run, it provides more detailed information about outbreaks, including the ability to capture heterogeneity in mixing and interactions. Hunter et al. call for further work to expand the ABM for complex scenarios and apply it on real-world data.

Here we present an extension of this work, by introducing a Monte-Carlo, Agent-based Model (MAM), a novel method to model infections and pandemic spread. Similarly to Hunter et al. [9], each agent represents an individual that changes position daily. Our method uses a Monte-Carlo algorithm to simulate daily position changes, allowing to model the spread of the disease using clusters of particles. In MAM, the probability of infection is a function of the distance between the two particles; therefore, a dense cluster has a higher infection probability [24,25]. These characteristics provide MAM with high flexibility in modeling pandemic spread. With MAM, we can model situations that consider not only the vaccination rate of the population but also changes in vaccine effectiveness over time, such as waning immunity. Furthermore, the flexibility of MAM enables us to model multiple variants simultaneously, each with a different infection rate.

We apply MAM to three major outbreaks of COVID-19 in Israel, which were dominated by different variants and different social behaviors. We show that MAM provides more flexibility and outperforms SIR-based models in predicting the spread of infectious diseases in heterogeneous populations.

2. Methods

2.1. COVID-19 infection rate definition

For modeling the spread of a pandemic, the most essential input required to define is R, the reproduction rate; when R is above 1, one individual infects, on average, more than one other individual, which indicates the disease is spreading. We distinguish between R_0, the basic reproduction number; R_e, the observed reproduction rate in the population; and R_t, the theoretical reproduction rate of the disease. Of those, R_e is affected by many factors that determine infectiousness, including NPIs and vaccinations, which vary by age groups. Testing policies, which vary across countries over time, may also impact the assessment of R_e. Thus, relying solely on daily case numbers as a measure of the disease spread is problematic, especially when comparing disease spreads among different countries [26]. In contrast to R_e, R_t is an estimation of the rate of encounters between infected and non-infected individuals that would have resulted in an infection without vaccinations. In MAM, R_t is the particle density or the size of the area for the particles (A), which is similar for all age groups.

2.2. The Monte Carlo Agent-based model (MAM)

Here we provide a short description of MAM and a pseudocode for the algorithm is available in the **Supplementary Methods**. The MAM

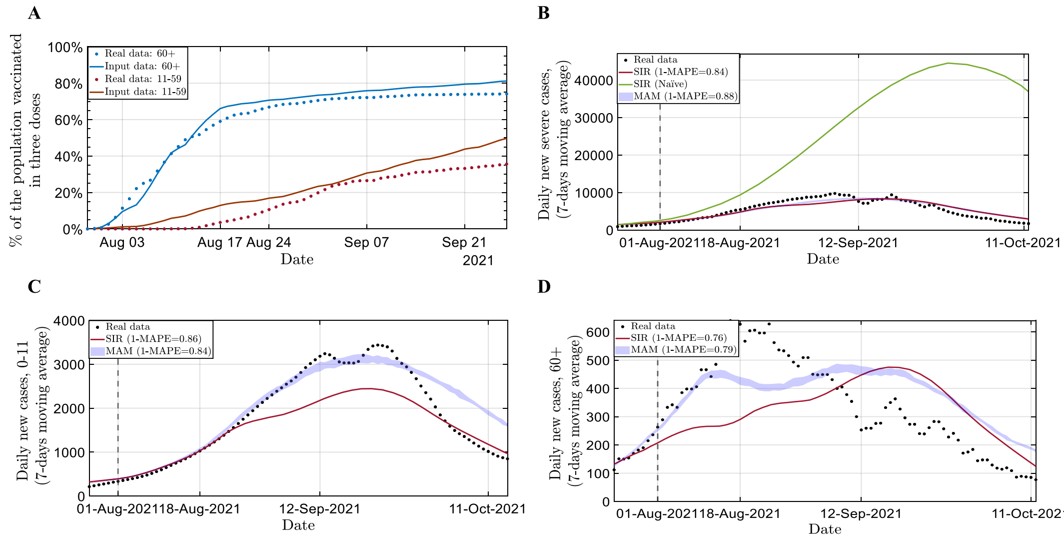

Fig. 2. Modelling the outbreak spread in Israel during the booster vaccination campaign. A. Vaccination rates of the booster dose for two age groups based on real data and predictions from August 1st' 2021. Actual vaccination rates are in dots. **B-D**. Predictions for the number of confirmed cases in Israel from July 24st, 2021, until Oct 15, 2021 (7-days moving average) for the whole population (**B**), ages 0–11, which are all unvaccinated at this time (**C**); and for 60 + years old (**D**). For panels B-D: real data in dots, SIR predictions in solid red line and MAM predictions in blue band (with 95% prediction interval). Dashed black line on August 1st, 2021, shows the index date from which predictions begin. Green line in **B** is a prediction made with SIR without breaking down age groups. MAPE statistics are presented in the legend. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

approach is to generate a spatial model of the population by using a set of interacting classical particles, as in an agent-based model. We then apply standard Monte Carlo (MC) procedures of sampling the transition among subsequent states [27-29].

Within the model, each particle can be in one of four states: (i) susceptible and unprotected (unvaccinated), (ii) susceptible and protected (vaccinated), (iii) currently infected and contagious, and (iv) recovered/dead. On any given day, the probability of a susceptible particle becoming infected depends on how far it is from every contagious particle around it, and whether the particle is protected. Particles change their position every day using an MC procedure. Since the infection probability is a function of the relative distance between particles/individuals, we can use different baseline spatial characteristics, that may represent a scattered city suburb or a dense university classroom. Moreover, the spatial features of the model allow simultaneous modeling of multiple geographic areas, each of which has unique vaccination rates and allows modeling infection for different infection circuits, such as families and close and remote communities [30].

2.3. Modelling COVID-19 spreads in Israel for three outbreaks using two different models

This subsection will discuss the differences between modeling outbreaks using SIR and MAM. Note that SIR has an adaptation, SEIR (Susceptible-Exposed-Infected-Removed/Recovered), that may be more suitable for modelling outbreaks with vaccinations. However, in this work we demonstrate how the SIR model can be adapted to model both unvaccinated and vaccinated populations simultaneously by adjusting the infection rate of the vaccinated population following its vaccination rate.

To determine the rate of pandemic spread for each outbreak, we need to evaluate both R_t and R_e. The actual spread rate of the disease is measured by R_e, which reflects both the protection of the population as a result of vaccinations and other characteristics, such as infectiousness of the dominant variant and NPIs, while R_t represents the spread of disease without vaccinations.

The main difference between the SIR and MAM is that in MAM each particle can be tuned individually, as opposed to SIR, where there are limited number of subgroups that can be modeled together, and each group has its own rate, R_e. In MAM, R_t, which is a function of the level of infectiousness of the variant (R_0) and the current NPIs, serves as input for the simulation. Hence, vaccination rates and vaccine effectiveness (VE), can be tuned for as many subgroups as needed. Therefore, this modeling approach offers high flexibility for modelling spread in populations that have different dynamics in age groups, geographical areas, etc., and in adapting the model to different VE scenarios.

For each outbreak, we estimate R_t, which is a dynamic metric, through relaxing and stringing restrictions, and the introduction of new variants (Fig. S1**A**). The Alpha variant, which was dominant in Israel in the beginning of 2021, had an R_0 of approximately 4, while the Delta variant, which became prevalent in Israel at the beginning of May 2021, had an R_0 of 5–8 (Fig. S1**B**) [31]. It should be noted that testing-availability and testing-policy vary not only by country but also by time (Fig. S1**C**), and thus, confirmed cases is a problematic metric for comparing disease spread across countries and across time. In this work, we estimate R_t in Israel from January 2021 to February 2021 (the first outbreak) to be 1.1–1.2, similarly to the official estimates of R_t in Israel at the same time [32]. For the case of the delta and omicron outbreaks, because of the removal of most social restrictions in Israel, we estimate R_t for the Delta and Omicron outbreaks to be twice than in the first outbreak, i.e., R_t= 2.2–2.5 (see **Table S1 and S2**).

Also, for both models, we need to have an estimation of the individual VE which might be non-trivial as it is also a dynamic measure that changes over time. Here, VE was set to 0% on days 0–7 after the first dose of vaccination and increases linearly up to 90% on the next 21 days, which is also one week after the second dose (Fig. S2**A**) [33-35]. We next assume reduction to 60% 180 days later [36], and following the booster dose (third dose), we assume increase back to 90% seven days after the vaccination [37]. In MAM, VE is provided as input for the model, while in SIR, VE is the input via R_e.

Vaccination rates differ among different populations and age groups (Fig. S2**B**). In Israel, by the end of 2020, in parallel with a major COVID-19 outbreak, a massive vaccination campaign was initiated. Elderly individuals were offered to vaccinate first, and every couple of weeks the age limit was reduced. By early January 2021, over 60% of those 60 years old and over were vaccinated with at least one dose. It was later offered to younger individuals to vaccinate, but this was a much slower process. This exemplifies, for the SIR model, R_e cannot be tuned

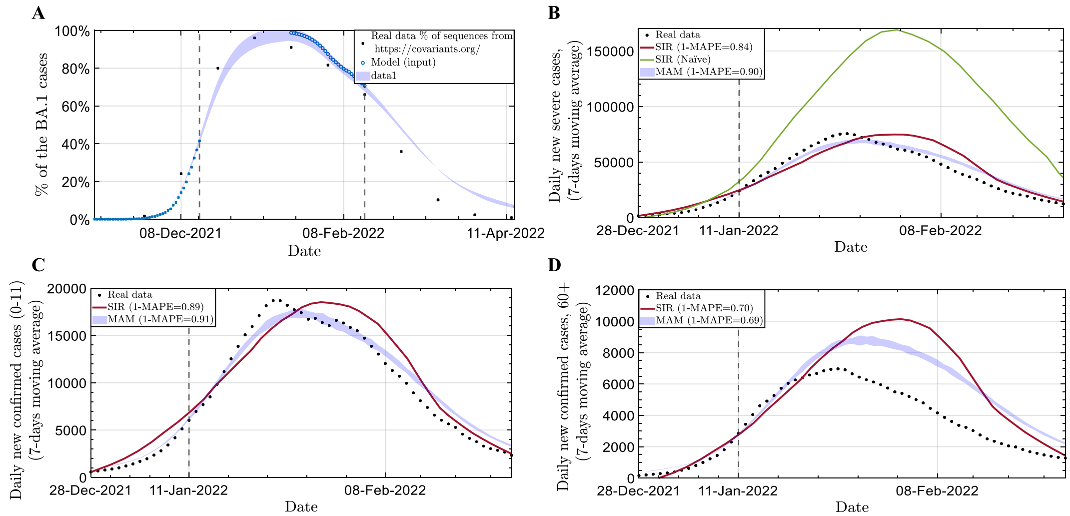

Fig. 3. Modelling the outbreak spread in Israel during the Omicron outbreak. A. Percentages of BA.1 variant from December 1st, 2021 until April 11th. Prediction with MAM for two time periods: from December 15th' 2021 up to January 18th, as it replaced the dominate Delta variant and from February 15th, 2022, as it was replaced by the BA.2 variant. Black dots are actual data.[42] and the blue circles are the model input (for each time period we used a dashed line to shows the index date from which predictions begin). **B-D**. Predictions for the number of confirmed cases in Israel from January 1st, 2022, until February 28, 2022 (7-days moving average) for the whole population (**B**), ages 0–11, which are all unvaccinated at that time (**B**); for 60 + years old (**D**). For panels B-D: real data in dots, SIR predictions in solid red line and MAM predictions in blue band (with 95% prediction interval). Green line in **B** is a prediction made with SIR without breaking down age groups. MAPE statistics are presented in the legend. Dashed black line on January t 11th, 2022, shows the index date from which predictions begin. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

correctly to the whole population but is a measurement that is specific to subgroups of the population.

In this work, for modeling the spread, we first used a "naïve" SIR model, which assumes homogeneous protection for all the population without age-based divisions. Next, we used a multi-age SIR [38,39] and MAM models such that we divided the population into three age groups: Group A represents the 0–11 years old population, which is 20% of the population who was not vaccinated until the end of 2021; Group B represents the 12–59 years old population, which is 63% of the population; and Group C which represents the 60 years old and over population, which is 17% of the population. The data of the ages distribution in Israel is based on official data from Central Bureau of Statistics in Israel. This data is also available at *https://github.com/hdeleon1/MAM—Flexible-Monte-Carlo-Agent-based-Model-for-Modelling-COVID-19-Spread/*.

To exemplify how this affects modeling, assume a theoretical case that the entire age group is vaccinated on Feb 1st. We would expect that after 28 days, the average protection against infection of that age group would be 90% (Fig. S2**B**), and therefore, their chance of infection is 10%. As a result, effective reproduction rates should vary for each group based on vaccination rates (Fig. S2**C-D**), despite the assumption that populations mix homogeneously, and everyone is susceptible to infection without vaccinations (Fig. S2**C-D**). Therefore, we have a system of 9 coupled differential equations (3 health status for each of three age groups) which we need to solve simultaneously (**Supplementary Data 2**). All parameters used for the calculations of the outbreaks for both the SIR and MAM models, are provided in **Supplementary Methods, in Tables S1-3**.

2.4. Statistical analysis

To provide statistical measurement for the accuracy of the modeling predictions compared to the actual real-world data, we used the mean absolute percentage error (MAPE), such that the accuracy is 1-the mean absolute percentage error [40]. This relative error measure uses absolute values to prevent positive and negative errors from canceling each other out and uses relative errors to compare forecast accuracy between time-series models. For each model, we examine the accuracy of the prediction in three age groups: the total population, the youngest and unvaccinated population, and the elderly population.

2.5. Error estimation

MAM uses the Monte Carlo algorithm, so the values shown are the average and standard deviation obtained by performing the analysis more than 800 times. The plots show the 95% prediction interval. Note that this 95% prediction interval is for the presentation purposes only; for calculation of MAPE, only the average value was used.

2.6. Data

All the data that was used in this work was taken from public data sources. Daily confirmed cases in Israel was taken from https://datadashboard.health.gov.il/COVID-19/general [41] and vaccination rates were taken from https://data.gov.il/dataset/covid-19 [42], all stratified by age groups and available from the Ministry of Health from December 2020 to February 2022. In Addition, for the Omicron outbreak, we used data from https://covariants.org/ [43] to examine the prediction for variants distribution over time. The data and the code used for this work is available at *https://github.com/hdeleon1/MAM—Flexible-Monte-Carlo-Agent-based-Model-for-Modelling-COVID-19-Spread/*.

3. Results

3.1. Outbreak 1: First vaccination campaign in Israel (Alpha variant)

We present the predictions of SIR and MAM modelling of the COVID-19 outbreak in Israel that started in mid-December 2020, parallel to the vaccination campaign. The Alpha variant dominated this outbreak. Data on confirmed cases and vaccination rates were collected until January 8th, 2021 (dashed horizontal line in Fig. 1), and predictions were made from this day forward. Vaccination rates from this date forward were estimated using an exponential function (Fig. 1**A**).

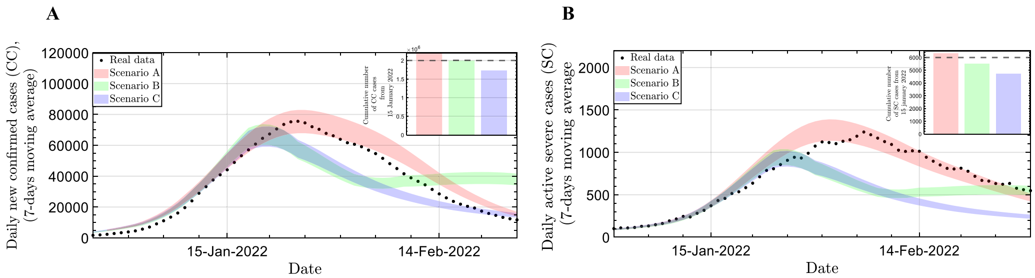

Fig. 4. Comparing effects of alternative scenarios of enforcing restrictions. A-B. Comparison of predictions for three scenarios for the daily number of confirmed cases (**A**) and parallel severe cases (**B**) in Israel from December 28th, 2022, until February 28th, 2022 (7-days moving average) for the whole population. Scenario A is no change in R_t, scenario B is temporary 20% reduction in R_t for two week and scenario C is 20% reduction in R_t. Dashed black line on January 11th, 2022, shows the index date from which predictions begin. For both panels, the inner plot is the cumulative number of cases from Jan 15th t until Feb 28th for the three scenarios. The dashed line is the actual data taken from [41].

First, we apply a naïve SIR model, assuming average protection of the total population (Fig. 1**B**). Interestingly, a naïve SIR approach fails and overestimates cases since the average protection of the total population is lower than the average protection for each vaccinated individual, as it considers the unvaccinated. This causes the total R_e to be overestimated, and in turn overestimate the number of expected confirmed cases.

We, therefore, applied multi-age SIR and MAM models to predict the number of confirmed cases from January 9th to mid-February 2021 in three age groups (Fig. 1**B-D**). Both SIR and MAM models were able to accurately predict the peak, the date when the number of confirmed cases peaked and started to decline ($R_e < 1$). However, MAM consistently outperformed SIR in predicting the decline in all three predictions. Across all three analyses, MAM outperformed SIR in terms of accuracy (MAPE = 89% for SIR vs. 95% for MAM).

3.2. Outbreak 2: Waning immunity and booster dose (Delta variant)

This outbreak, which started in June 2021, was characterized by three factors that overlapped: relaxation of NPIs, the introduction of the more infectious Delta variant, and waning immunity half a year after the first vaccination campaign. The combination of all the above resulted in a higher R_t than that of the first outbreak [44]. On July 31, 2021, Israel initiated another vaccination campaign for a booster dose, which similarly took place to the first campaign: the elderly population first, and later opened it for the rest of the population.

Here, we modeled the outbreak according to information gathered until August 1st, 2021 (dashed line in Fig. 2), as the vaccination campaign has just started. Since vaccination dynamics were unknown then, we assumed similar dynamics, by age groups, as in the first vaccination campaign (Fig. 2**A**). We first applied the naïve SIR model to the whole population, assuming only the average protection for the whole population (Fig. 2**B**). As a result of the assumption that the unvaccinated population has a higher R_e than that of the first outbreak, the number of daily confirmed cases was significantly overestimated. This significant deviation results from the exponential nature of pandemic spread, a 10% increase in R_e can result in a 50% increase in the number of total cases. We conclude that a naïve model cannot be applied to accurately predict the spread in scenarios where the infection potential for the unvaccinated is significantly higher than for the vaccinated.

Both multi-age SIR and MAM models were able to predict the dynamics of this outbreak well, yet MAM predicted the peak better and had an overall accuracy of 88% compared to 84% with SIR (Fig. 2**B**). At the young age group (ages 0–11), MAM accurately predicted the peak, but compared to the real data, the decrease of this outbreak was much sharper than predicted (Fig. 2**C**). A possible explanation is that the peak of the outbreak intersected with the Jewish high-holidays, which is a time of school closing, and later, a program of mandatory rapid testing before return to school, and both together made the outbreak subside quickly in young children. Both models did not perform well in the age group of 60 + years old, but both had relatively good estimation for the end of this outbreak. Like all other comparisons, MAM was more accurate than SIR here as well (Fig. 2**D**).

3.3. Outbreak 3: Competing variants

This outbreak, which started in December 2021, was characterized by the highly infectious Omicron variant. This variant rapidly became dominant, with individuals arriving from abroad with Omicron, while Delta was also on the rise. Where Delta and Omicron differ significantly in their R_0: here we assume that Omicron R_0 is double of Delta's R_0 and in addition, we assume a shorter incubation time for Omicron (Table S3. A similar scenario happened later in February, where Omicron BA.1 was replaced by the BA.2 variant, again because of higher R_0 (we assume here 50% higher R_0 for BA.2 compared to BA.1) [45-47]. Using data on number of infected individuals that arrived in Israel in the first half of December (assuming all are Omicron) and number of confirmed cases in Israel in that time frame (assuming all are Delta), we attempted to predict the proportion of the variants in the next 30 days. By using MAM modelling, where each particle represents an individual, such a task is feasible, and MAM was able to accurately predict the dynamics of variants transition. We also repeated this modelling for the BA.1-BA.2 transition, using data of influx from January 19, 2022, to the mid of February, and predicted variants distributions in February-mid till April 2022 (Fig. 3**A**).

We next used data collected up to January 11, 2022, to model the outbreak from then forward, for both SIR and MAM models, and predict daily confirmed cases under the assumption that from that day and on, Omicron is the dominant variant, which enable the use of the multi-age SIR model. Based on the data gathered up to the index date, we assumed here $R_t = 2.5$. In MAM, since there are competing variants, we also assumed that Omicron is twice as infectious as Delta.

Like the previous outbreaks, we first employ the naïve SIR model (Fig. 3**B**), which again overestimated the actual data, but not extreme overestimation as in in outbreak 2. This is because in this outbreak the vaccinations provided only little protection. We next applied MAM and multi-age SIR to the three age groups. Both models successfully predicted the number of cases in the whole population and the young populations, with MAM predicting better the peak and with better MAPE (Fig. 3**B-D**). However, both models did not predict well the peak and the rapid decrease in cases in the 60 + year old population (Fig. 3**D**), which may be a result of another vaccination campaign, this time for a fourth dose, and this time only for individuals over 60 years old. By January 10, 2022, 15% of the 60 + year-old population received the fourth does, and two weeks later, this figure went up to about a quarter. Based on the deviation between our predictions and the number of daily confirmed cases, which begun after January 18th, we hypothesized that this vaccination campaign was the primary reason for the early drop in cases. However, is yet a debate of how effective this campaign was and whether it had an effect and the outbreak [48,49]. Also, other possible explanation for this deviation between the prediction and the actual data is that the older population have taken extra precautions during a major outbreak.

Finally, an important question is how changes in R_t, possibly by enforcement of more stringent NPIs, may affect the dynamics of the outbreaks. We used MAM to compare how a reduction of 20% in R_t will affect the number of daily confirmed cases and, possibly more importantly, the number of daily active severe cases in hospitals in parallel, where number of daily active severe cases was calculated using the daily number of confirmed cases, assuming a 0.25% chance of a confirmed case to become a severe case with an average hospitalization time of 6 days [50]. This was the primary metric used by the government to decide whether enforcing stringent NPIs is required, as the main goal was to prevent hospital overflow. We compared three scenarios: A) no change in R_t; B) Temporary reduction in R_t for two weeks starting January 15, which may reflect two-weeks school closure and C) Reduction of R_t by 20% from January 15 (Fig. 4**A-B**). For scenario C, the analyses predicted 40% fewer total infections and severe cases from January 15 until February 28 in comparison to scenario A (Fig. 4**A-B**). Importantly, in both scenarios B and C, our prediction suggested that the peak of active severe cases would not have crossed 900 severe cases in parallel (Fig. 4**B**), which was the limit in Israel before a significant decrease in the level of care [51].

Another insight from these analyses is the divergence that we observe between the actual data and predictions from around January 24. We only observe sharp reduction compared to the prediction in confirmed cases, but not in severe cases. During that time, testing policy in Israel changed, and there was arguably undercount of number of actual cases. However, since the number of severe cases is not affected by changes in testing policies, the accuracy of the model to predict severe cases was not harmed, suggesting that such an approach can be used to estimate undercounting due to policy changes.

4. Discussion

As SARS-CoV-2 evolves, its realities are rapidly changing with the introduction of numerous variants, each with its own characteristics, vaccination campaigns, waning immunity, and constantly changing NPIs and testing requirements. It is, therefore, necessary to develop a dynamic model capable of taking all these factors into account when modeling disease dynamics. In this paper, we present a novel particle-based approach for modeling disease outbreaks, known as MAM, that provides a high level of flexibility for modeling such complex situations. We showed its superiority over the widely used multi-age SIR model in predicting the dynamics of outbreaks in Israel; all analyses used similar assumptions for both models. Moreover, we demonstrated that MAM is highly flexible, allowing us to easily define different vaccination rates in subgroups of the population, changes in NPIs, and competing variants.

We found that a naïve SIR model significantly overestimates outbreaks, especially when vaccines are in effect. This could be by the Simpson paradox, as vaccines are effective on the individual level, but when merging all age groups together, which have different vaccination rates, the average protective effect seems low. Also, unvaccinated individuals also benefit from indirect protection when vaccines provide high levels of protection [52]. Because vaccination campaigns in Israel were conducted according to age groups, there was cumulative protection in groups that were vaccinated early in addition to the vaccine-induced protection. Due to this, assuming uniform protection throughout the population is not sufficient to accurately model the disease's spread. Therefore, we applied a multi-age approach using SIR and MAM. In this approach, each age group has its own characteristics based on vaccine effectiveness and vaccination rate and assuming homogenous mixing across the three age groups. Both approaches provided relatively good predictions in the three outbreaks we modeled from Israel; however, MAM was consistently more accurate. It is important to note that here we used only three age groups and not more, as adding additional age groups to SIR is complex and prawn to errors. In MAM, this is straightforward.

Another complication we introduced here is modelling outbreaks when there are competing variants. We showed here that such situations can be modeled efficiently from the very early stage of the outbreak with MAM, since we can provide each particle different adhesion characteristics, such as adhesion potential and incubation time. Also, dynamics of infection can vary across different geographies. The MAM approach has a built-in spatial structure, thus, adding spatial characteristic, such as households, neighborhoods, cities etc. can easily be applied to this model, and provide additional precision to modelling the disease spread [30].

In summary, we presented here a novel approach for modelling the spread of the pandemic and exemplified its accuracy and flexibility across different scenarios. We hope that this tool will be utilized in the future to make informed decision-making on how to deal with disease outbreaks.

CRediT authorship contribution statement

Hilla De-Leon: Conceptualization, Methodology, Data curation, Software, Writing – original draft. **Dvir Aran:** Supervision, Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbi.2023.104364.

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